

Title: Smart Review Analyzer for Multi-Platform User Feedback

Team Details:

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Problem Statement: In the digital era, user-generated content across e-commerce sites, video platforms, and social media has become a voluminous source of feedback. Businesses and content creators face a significant challenge in manually processing this data, leading to a delayed understanding of customer satisfaction and a failure to grasp the specific emotions driving user opinions. This results in missed opportunities for empathetic engagement, targeted product improvement, and a reactive rather than proactive approach to market trends. This project addresses the critical need for an automated system to analyze and act upon multi-platform user feedback effectively.

Objectives: The primary objectives of this project are:

1. To develop an automated system for collecting user reviews and comments from diverse digital sources.
2. To pre-process and structure the heterogeneous raw data into a unified, analyzable dataset.
3. To implement Natural Language Processing (NLP) models for accurate sentiment and emotion classification (e.g., positive, negative, joy, anger, frustration).
4. To develop a workflow to automatically prioritize and flag reviews based on detected emotional intensity for immediate attention.
5. To identify significant themes and topics discussed within the feedback.
6. To design an intuitive dashboard to visualize the analytical results for actionable insights.

Proposed Methodology/Approach: The proposed system will utilize web scraping and APIs to harvest user reviews. The collected data will undergo extensive pre-processing, including cleaning and normalization. A transformer-based architecture (e.g., BERT) will be fine-tuned on a dataset labeled for both sentiment and discrete emotions. Following classification, a

rules-based engine will process the emotional output to flag and prioritize reviews indicating high urgency (e.g., intense anger or disappointment). Topic modeling algorithms like Latent Dirichlet Allocation (LDA) will be employed to discover underlying themes. The entire system will be presented through an interactive web-based dashboard.

Expected Outcome: The expected outcome is a functional prototype of the "Smart Review Analyzer" that delivers not just an overview of user sentiment but also deeper emotional insights. The system will provide automated sentiment/emotion scoring and generate a prioritized queue of reviews requiring immediate attention. The final dashboard will present these insights through clear visualizations, such as trend graphs and word clouds, enabling stakeholders to make swift, data-driven, and empathetic decisions to enhance products, services, and brand reputation by attending directly to the most emotionally charged feedback.

Base Research Paper Reference: Rana, T. A., & Cheah, Y. N. (2022). A systematic review on aspect-based sentiment analysis: Resources, techniques, and challenges. *Artificial Intelligence Review*, 56(8), 8139–8193.

<https://link.springer.com/article/10.1007/s10462-024-10906-z>